

TECHNICAL SYSTEM BLUEPRINT



Developer & User System Blueprint

A detailed breakdown of every component, trigger logic, what happens when pressed, and what gets affected.

Prepared for: THOLA Platform Product Handover & System Review
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1. Entrance & Trust Portals

[index.html](#) | [learn-more.html](#)

Exposes the marketplace landing views, controls theme preferences, and describes safety layers.

LANDING PAGE TRIGGERS & ELEMENTS

Theme Toggle Button (`#theme-toggle`)

UI CONTROLLER

Sleek Sun/Moon icon toggle inside the main navigation bar.

If you press: Triggers the `toggleTheme()` script in `js/main.js`.

What gets affected: Swaps the HTML attribute `data-theme` between "light" and "dark". Saves selection to `localStorage.setItem('theme', 'dark')` so preferences persist across pages.

"Get Started" / "Join Today" Action Targets

LINK TRIGGER

Primary orange accented call-to-action buttons in the hero panel.

If you press: Triggers an immediate browser window redirection.

What gets affected: Navigates browser state to `register.html`.

LEARN MORE PAGE TRIGGERS

Safety Feature Cards (Vouch, Verification, Escrow)

INFORMATION BLOCKS

Visual cards outlining the trust ecosystem.

If you press: These blocks are non-interactive informative assets.

What gets affected: Boosts user confidence and explains trust metrics.

2. User Authentication Suite

login.html | register.html

Handles local authentication and user profiles secure initialization using hashed keys.

REGISTRATION FORM ELEMENTS

Password Input Fields

FORM INPUT

Accepts user password string with match validation.

If you type: Verifies that values in both Password and Confirm Password inputs match.

What gets affected: If mismatched, disables the submit trigger and displays validation warnings in the UI.

"Create Account" Button

FORM SUBMIT

Submits email, username, and password fields.

If you press: Intercepts form submission, hashes password string, and creates a user record object.

What gets affected: Writes new profile records into `localStorage.setItem('users', JSON.stringify(...))`. Sets credentials state and redirects to `dashboard.html`.

LOGIN PAGE COMPONENTS

"Log In" Button

FORM SUBMIT

Checks login details against existing profiles database.

If you press: Validates entered email and password against stored keys in `localStorage`.

What gets affected: Updates `sessionStorage.setItem('loggedInUser', ...)` on success, launching dashboard. Displays error validation banners if mismatch occurs.

3. Marketplace Dashboard System

dashboard.html | js/dashboard.js

Handles dynamic filter evaluations and structures item grid layouts.

INTERACTIVE FILTER CONTROLS

Search Bar (`#searchInput`)

TEXT FILTER

Text input that monitors query string keywords in real-time.

If you type: Triggers immediate list filtering matching string inputs against listing titles and descriptions.

What gets affected: Renders matched items dynamically. Non-matched list items are hidden from the layout grid.

Category Select Cards (Skills / Goods / All)

DYNAMIC FILTER

Category buttons located above the main grid layout.

If you press: Updates target category state variable in memory and updates selection indicator class.

What gets affected: Instantly narrows listings grid down to matching item types (e.g. only service skills).

Province Dropdown Menu (`#provinceFilter`)

SELECT DROPDOWN

Dropdown selector containing all 9 South African provinces.

If you select: Evaluates selected province value against listing location fields.

What gets affected: Restricts grid items to locations matching the chosen province. If no items match, loads a placeholder empty-state graphic.

4. Service Details & Publisher

listing-detail.html | post-listing.html

Facilitates service details verification, contact handshake routing, and new post persistence.

LISTING DETAIL INTERACTIONS

"Contact Provider" Action Button

THREAD TRIGGER

Primary action button on detail sidebar.

If you press: Reads the profile username of the listing creator.

What gets affected: Caches chat partner detail in `sessionStorage.setItem('activeChatPartner', sellerName)` and directs browser session to the chat panel (`chat.html`).

POST LISTING PORTAL INPUTS

Province Location Selector

DROPDOWN SELECT

Standardized dropdown selector listing all 9 South African provinces.

If you select: Binds chosen province slug to the new listing's location attribute.

What gets affected: Establishes a clean location parameter, ensuring the listing maps correctly to dashboard search indexes.

"Publish Listing" Submission Button

FORM SUBMIT

Submits listing name, description, category, pricing, and image payload.

If you press: Prevents page reload, parses input data fields, and constructs a new listing object.

What gets affected: Appends new listing item data into the active session database, shows a verification notification, and redirects window focus back to `dashboard.html`.

5. Communication & Identity Services

[chat.html](#) | [profile.html](#)

Drives the messaging engine and controls biometric/ID verification validation badge rendering.

CHAT INTERFACE TRIGGERS

Contact Selector Sidebar

THREAD SWITCHER

Lists usernames representing all active communication threads.

If you press: Switches context to show message history related to the selected user.

What gets affected: Renders active messages log and highlights target name status inside the chat interface.

Send Button (`#btnSend`) / Enter Key Press

MESSAGE DISPATCH

Submits input message text.

If you press: Packages text content with a dynamic timestamp and user metadata.

What gets affected: Appends chat record into `localStorage` message database and triggers a live update of the conversation viewport.

PROFILE HUB VERIFICATION TRIGGERS

"Verify Identity" Button

VALIDATION TRIGGER

Uploads document image for ID validation.

If you press: Processes mock validation checks.

What gets affected: Updates user status in database setting `verifiedID: true`. Renders green check badge next to the user's name across all system views.

6. Technical Operation Lifecycle

[System Data Flow Map](#)

Step-by-step technical pipeline of a transaction lifecycle and state persistence.

1

Listing Initialization

Sellers fill attributes in `post-listing.html`. Pressing **Submit** serializes the input values into a JSON object and updates the database cache in `sessionStorage`.

2

Dashboard Indexing

`dashboard.js` reads the listing array, executes filter calculations matching searches, province dropdown matches, and updates grid elements in the DOM.

3

P2P Thread Negotiation

Pressing **Contact Provider** saves listing identity info into temporary storage and loads `chat.html`. The chat module loads matched communication logs from `localStorage`.

4

Trust Accumulation

Submitting a vouch on a seller profile increases their Vouch Count. This database value is saved to `localStorage` and dynamically appears on all of their future marketplace listing cards.